DATA PREPARATION AND EXPLORATORY DATA ANALYSIS

3. Data Cleaning and Preparation

3.1 Understanding the Dataset

The first step was to examine the structure of the Excel workbook and understand the purpose of each worksheet. The workbook contains five tables representing different types of racing information.

The race_results worksheet contains driver-race records, while the lap_intelligence worksheet contains individual lap records. Qualifying results, championship standings, and circuit information are stored separately.

Understanding these differences is important because each worksheet represents a different level of detail.

3.2 Data Preparation Process

The preparation and analysis workflow involves the following activities:

1. Reviewing the worksheets and their column names.
2. Identifying numerical, categorical, and other relevant variables.
3. Checking for missing values in important fields.
4. Selecting relevant measures and dimensions for analysis.
5. Organizing the data according to the requirements of each visualization.
6. Ensuring that charts use appropriate aggregations and levels of detail.

Any additional cleaning, filtering, or transformation should be described according to the operations actually performed during the project.

3.3 Data Quality Considerations

The supplied dataset contains missing values in several fields.

In the race_results worksheet:

- finish_position has 396 missing values.
- positions_gained has 396 missing values.
- gap_to_leader_s has 396 missing values.

- total_race_time_s has 6,130 missing values.

These missing values may affect analyses involving finishing positions, position changes, and race completion times.

The workbook also contains some values that do not consistently match official historical Formula 1 records. Therefore, the findings in this project should be interpreted as results from the supplied dataset rather than as verified official historical statistics.

4. Exploratory Data Analysis

4.1 Introduction

Exploratory Data Analysis (EDA) involves examining a dataset to understand its structure, summarize important variables, and identify patterns.

In this project, EDA was used to examine race results, lap performance, qualifying results, championship standings, and circuit information. This helped identify useful statistics for the Tableau visualizations.

4.2 Race Results Analysis

The race_results worksheet contains 6,464 records and 334 distinct race IDs.

The following statistics were calculated from the supplied data:

- Total recorded pit stops: 11,464.
- Mean pit-stop count: approximately 1.77 per driver-race record.
- Total recorded DRS overtakes: 9,123.
- Total recorded laps led: 24,631.
- DNF-marked records: 396, representing approximately 6.13% of race-result records.

These statistics provide a basis for exploring race outcomes, pit-stop activity, overtaking, and driver performance.

4.3 Lap Intelligence Analysis

The lap_intelligence worksheet contains 3,814 records, including lap times, sector times, tyre compounds, tyre age, DRS activity, and speed measurements.

The highest recorded speed-trap value is 379.4 km/h, while the lowest recorded lap-time value is 66.587 seconds.

The tyre-compound distribution is:

Tyre Compound	Lap Records
Soft	1,202
Medium	1,174
Hard	1,158
Intermediate	280
Total	3,814

Soft tyres account for approximately 31.5% of the lap records, while Intermediate tyres account for approximately 7.3%.

These figures represent tyre-compound labels in the supplied lap records, not the number of unique tyre sets used in races.

4.4 Qualifying Analysis

The qualifying_results worksheet contains 3,130 records.

It includes qualifying positions, qualifying lap times, gaps to pole position, sector times, and speed measurements.

These variables can be used to compare qualifying performance, examine differences between drivers, and explore the relationship between qualifying positions and other recorded performance measures.

4.5 Championship Standings Analysis

The championship_standings worksheet contains 311 records relating to driver standings and championship performance.

The highest cumulative point totals, calculated by summing the points field across each driver's records, are:

Rank	Driver	Cumulative Points
1	Max Verstappen	4,640
2	Lewis Hamilton	4,491
3	Sergio Perez	3,366

4	George Russell	3,299
5	Charles Leclerc	3,243

These totals represent sums across the included records, not points from a single season. They should be interpreted as statistics from the supplied dataset.

4.6 Weather Analysis

The weather distribution, counting each distinct race ID once, is shown below.

Weather Condition	Number of Races	Percentage
Dry	264	79.0%
Overcast	27	8.1%
Light Rain	19	5.7%
Heavy Rain	12	3.6%
Mixed	12	3.6%
Total	334	100%

Dry is the most frequently recorded weather category in the dataset.

This analysis demonstrates how categorical information can be summarized using counts and percentages and represented through bar charts or pie charts.